

[Homework 3] Martingale and Stopping Time

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Problem 1 (A maximal inequality)

(a) The definition of a martingale tells us

$$\mathbf{E}[Z_{k+1} \mid \mathcal{F}_k] = Z_k \quad (1)$$

Consider squared expansion,

$$\begin{aligned} (Z_k - Z_{k-1})^2 &= Z_k^2 - 2Z_k Z_{k-1} + Z_{k-1}^2 \\ \mathbf{E}[(Z_k - Z_{k-1})^2 \mid \mathcal{F}_{k-1}] &= \mathbf{E}[Z_k^2 \mid \mathcal{F}_{k-1}] - 2\mathbf{E}[Z_k Z_{k-1} \mid \mathcal{F}_{k-1}] + \mathbf{E}[Z_{k-1}^2 \mid \mathcal{F}_{k-1}] \\ &= \mathbf{E}[Z_k^2 \mid \mathcal{F}_{k-1}] - Z_{k-1} \mathbf{E}[Z_k \mid \mathcal{F}_{k-1}] + Z_{k-1}^2 \\ &= \mathbf{E}[Z_k^2 \mid \mathcal{F}_{k-1}] - Z_{k-1}^2 \\ \mathbf{E}[\mathbf{E}[(Z_k - Z_{k-1})^2 \mid \mathcal{F}_{k-1}]] &= \mathbf{E}[\mathbf{E}[Z_k^2 \mid \mathcal{F}_{k-1}] - Z_{k-1}^2] \\ \mathbf{E}[(Z_k - Z_{k-1})^2] &= \mathbf{E}[Z_k^2] - \mathbf{E}[Z_{k-1}^2] \end{aligned} \quad (2)$$

and obviously

$$\sum_{k=1}^n \mathbf{E}[(Z_k - Z_{k-1})^2] = \mathbf{E}[Z_n^2] - \mathbf{E}[Z_0^2] \quad (3)$$

(b) When $t < \tau$, we know $Z'_t = Z_t$, $\mathbf{E}[Z'_{t+1} \mid \mathcal{F}_t] = Z'_t$. We need to prove that for $t \geq \tau$, we also have $\mathbf{E}[Z'_{t+1} \mid \mathcal{F}_t] = Z'_t$. This is actually $\mathbf{E}[Z_\tau \mid \mathcal{F}_t] = Z_\tau$, which is naive because we actually decide Z_τ on $\mathcal{F}_t$, or to say, Z_τ is $\mathcal{F}_t$ -measurable.

(c) First S_i is a martingale. Since $\mathbf{E}[X_i] = 0$,

$$\begin{aligned} \mathbf{E}[S_{i+1} \mid X_1 \dots X_i] &= \mathbf{E}[S_i + X_{i+1} \mid X_1 \dots X_i] \\ &= \mathbf{E}[S_i \mid X_1 \dots X_i] + \mathbf{E}[X_{i+1} \mid X_1 \dots X_i] \\ &= S_i + \mathbf{E}[X_{i+1}] \\ &= S_i \end{aligned} \quad (4)$$

Then we apply the result of question (a). We get

$$\Pr[\max_{1 \leq k \leq n} |S_k| \geq \lambda] \leq \frac{1}{\lambda^2} (\mathbf{E}[S_n^2] - \mathbf{E}[S_1^2]) \quad (5)$$

We shall define a martingale S' alike question (b). We define

$$S'_t = \begin{cases} S_t & \text{if } t < \tau; \\ S_\tau & \text{if } t \geq \tau. \end{cases} \quad (6)$$

by denoting $\tau = \arg \max_t |S_t|$. To prove the original statement we are to prove

$$\Pr[S'_\tau \geq \lambda] \leq \frac{1}{\lambda^2} (\mathbf{E}[S_n^2] - \mathbf{E}[S_1^2]) \leq \frac{1}{\lambda^2} (\mathbf{E}[S'^2_\tau] - \mathbf{E}[S'^2_1]) \quad (7)$$

However, $\mathbf{E}[S'^2_1] = \mathbf{E}[S^2_1] \geq 0$, $\mathbf{E}[S'_\tau] = \mathbf{E}[S'_0] = 0$

$$\Pr[S'_\tau \geq \lambda] \leq \frac{1}{\lambda^2} (\mathbf{E}[S'^2_\tau] - \mathbf{E}[S'^2_\tau]) = \frac{\mathbf{D}[S'_\tau]}{\lambda^2} \quad (8)$$

This is called the Chebyshev Inequality. No need to prove further.

Problem 2 (Biased random walk)

(a)

$$\begin{aligned}
 \mathbf{E}[S_{k+1} \mid X_1 \dots X_k] &= \mathbf{E}\left[\sum_{i=1}^{k+1} (X_i + 2p - 1) \mid X_1 \dots X_k\right] \\
 &= \mathbf{E}\left[\sum_{i=1}^k (X_i + 2p - 1) + X_{k+1} + 2p - 1 \mid X_1 \dots X_k\right] \\
 &= \mathbf{E}\left[\sum_{i=1}^k (X_i + 2p - 1) \mid X_1 \dots X_k\right] + \mathbf{E}[X_{k+1} \mid X_1 \dots X_k] + 2p - 1 \quad (9) \\
 &= S_k + \mathbf{E}[X_{k+1}] + 2p - 1 \\
 &= S_k + p \cdot (-1) + (1 - p) + 2p - 1 \\
 &= S_k
 \end{aligned}$$

(b)

$$\begin{aligned}
 \mathbf{E}[P_{k+1} \mid X_1 \dots X_k] &= \mathbf{E}\left[\left(\frac{p}{1-p}\right)^{Z_{k+1}} \mid X_1 \dots X_k\right] \\
 &= \mathbf{E}\left[\left(\frac{p}{1-p}\right)^{Z_k} \cdot \left(\frac{p}{1-p}\right)^{X_{k+1}} \mid X_1 \dots X_k\right] \\
 &= P_k \cdot \mathbf{E}\left[\left(\frac{p}{1-p}\right)^{X_{k+1}} \mid X_1 \dots X_k\right] \\
 &= P_k \cdot \mathbf{E}\left[\left(\frac{p}{1-p}\right)^{X_{k+1}}\right] \quad (10) \\
 &= P_k \cdot \left(p \cdot \frac{1-p}{p} + (1-p) \cdot \frac{p}{1-p}\right) \\
 &= P_k
 \end{aligned}$$

(c) For a time period with length $a+b$, there is always a possibility $p^{a+b} + (1-p)^{a+b}$. Therefore, if we divide the time into consecutive periods in this manner, in expected finite time, we can meet some period when the event happened. Hence, $\mathbf{E}[\tau] < \infty$. Moreover, $|P_{k+1} - P_k| = \left| \left(\frac{p}{1-p}\right)^{Z_{k+1}} - \left(\frac{p}{1-p}\right)^{Z_k} \right| < \max\left(\left(\frac{p}{1-p}\right)^{-a}, \left(\frac{p}{1-p}\right)^b\right)$ is bounded, so the third condition of OST holds, which implies that $\mathbf{E}[P_\tau] = \mathbf{E}[P_0]$, that is

$$\begin{aligned}
 \left(\frac{p}{1-p}\right)^{-a} \cdot P_a + \left(\frac{p}{1-p}\right)^b \cdot P_b &= 1 \\
 \begin{cases} P_a = \frac{1 - \left(\frac{p}{1-p}\right)^b}{\left(\frac{p}{1-p}\right)^{-a} - \frac{p}{1-p}} \\ P_b = 1 - P_a \end{cases} & \quad (11)
 \end{aligned}$$

Similarly, we have $|S_{k+1} - S_k| \leq 2$ is bounded, $\mathbf{E}[S_\tau] = \mathbf{E}[S_0]$, that is

$$\begin{aligned}
 P_a \cdot (-a) + P_b \cdot b + \mathbf{E}[\tau] \cdot (2p - 1) &= 0 \\
 \mathbf{E}[\tau] &= \frac{P_a \cdot a - P_b \cdot b}{2p - 1}, p \neq \frac{1}{2} \\
 \text{where } \begin{cases} P_a = \frac{1 - \left(\frac{p}{1-p}\right)^b}{\left(\frac{p}{1-p}\right)^{-a} - \frac{p}{1-p}} \\ P_b = 1 - P_a \end{cases} & \quad (12)
 \end{aligned}$$

when $p = \frac{1}{2}$, we've solved in class that $\mathbf{E}[\tau] = ab$.

Problem 3 (Learning theory)

- (a) Since $\sup_{h \in \mathcal{H}} |L(h) - L_S(h)| \leq \frac{\epsilon}{2}$, for global optimal h^* , $L_S(h^*) - L(h^*) \leq \frac{\epsilon}{2}$. For optimal on training set $\hat{h}$, h^* , $L(\hat{h}) - L_S(\hat{h}) \leq \frac{\epsilon}{2}$. Then,

$$\begin{aligned} L(\hat{h}) - \frac{\epsilon}{2} &\leq L_S(\hat{h}) \leq L_S(h^*) \leq L(h^*) + \frac{\epsilon}{2} \\ L(\hat{h}) &\leq L(h^*) + \epsilon \end{aligned} \quad (13)$$

- (b) We can see $\text{Rep}(S)$ is a function conditioned on $S_0, S_1 \dots S_n$ and we should define

$$Z_i \triangleq \mathbf{E}[\text{Rep}(S) \mid S_0 \dots S_i] \quad (14)$$

$\{Z_n\}_{n \geq 0}$ is a Doob Martingale w.r.t. $\{S_n\}_{n \geq 0}$ and for every $i \geq 1$, since S_i and S_{i-1} differ at most one sample,

$$\begin{aligned} \text{Rep}(S) &\triangleq \sup_{h \in \mathcal{H}} (L(h) - L_S(h)) \\ |Z_i - Z_{i-1}| &= |\mathbf{E}[L(h_1) - L_S(h_1) \mid S_0 \dots S_i] - \mathbf{E}[L(h_2) - L_S(h_2) \mid S_0 \dots S_{i-1}]| \\ &= |L(h_1) - L(h_2) - (\mathbf{E}[L_S(h_1) \mid S_0 \dots S_i] - \mathbf{E}[L_S(h_2) \mid S_0 \dots S_{i-1}])| \\ &\text{here } L(h_1) - L(h_2) \leq \frac{1}{m}, \\ &|\mathbf{E}[L_S(h_1) \mid S_0 \dots S_i] - \mathbf{E}[L_S(h_2) \mid S_0 \dots S_{i-1}]| \leq \frac{1}{m}, \\ &\text{and equation holds only once. In fact, if the equation doesn't hold, LHS=0} \\ |Z_i - Z_{i-1}| &\leq \frac{1}{m} \end{aligned} \quad (15)$$

Apply Azuma-hoeffding Inequality,

$$\mathbf{Pr}[|Z_m - Z_0| \geq t] \leq 2e^{-2mt^2} \quad (16)$$

let t be $\sqrt{\frac{1}{2m} \log \frac{2}{\delta}}$,

$$\mathbf{Pr}[|Z_m - Z_0| \geq \sqrt{\frac{1}{2m} \log \frac{2}{\delta}}] \leq \delta, \quad (17)$$

or, since Z_i is defined with sup,

$$\forall h \in \mathcal{H}, \mathbf{Pr}[L(h) - L_S(h) - \mathbf{E}_{S \sim \mathcal{X}^m}[\text{Rep}(S)] \leq \sqrt{\frac{1}{2m} \log \frac{2}{\delta}}] \geq 1 - \delta, \quad (18)$$

and given $\mathbf{E}_{S \sim \mathcal{X}^m}(S) \leq 2\mathbf{E}_{S \sim \mathcal{X}^m}[\text{R}(S)]$, finally

$$\forall h \in \mathcal{H}, \mathbf{Pr}[L(h) - L_S(h) \leq 2\mathbf{E}_{S \sim \mathcal{X}^m}[\text{R}(S)] + \sqrt{\frac{1}{2m} \log \frac{2}{\delta}}] \geq 1 - \delta, \quad (19)$$

- (c) First let t be $\sqrt{\frac{1}{2m} \log \frac{8}{\delta}}$ in (b) we get the conclusion

$$\mathbf{Pr}[L(\hat{h}) - L_S(\hat{h}) \leq 2\mathbf{E}_{S \sim \mathcal{X}^m}[\text{R}(S)] + \sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \geq 1 - \frac{\delta}{4}, \quad (20)$$

Similarly, we let $Y_i \triangleq \mathbf{E}[\text{R}(S) \mid S_0 \dots S_i]$ be a martingale. We still have

$$\begin{aligned} |Y_i - Y_{i-1}| &= \frac{1}{m} (\mathbf{E}[\sup_{h \in \mathcal{H}} \sum_{i=1}^m \sigma_i \cdot \mathbf{1}[h(x_i) \neq \ell(x_i)] \mid S_0 \dots S_i] - \\ &\quad \frac{1}{m} \mathbf{E}[\sup_{h \in \mathcal{H}} \sum_{i=1}^m \sigma_i \cdot \mathbf{1}[h(x_i) \neq \ell(x_i)] \mid S_0 \dots S_{i-1}]) \\ &\leq \frac{1}{m} \end{aligned} \quad (21)$$

Also similarly, let t be $\sqrt{\frac{1}{2m} \log \frac{8}{\delta}}$,

$$\Pr[|Y_m - Y_0| \geq \sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \leq \frac{\delta}{4}, \quad (22)$$

$$\Pr[\mathbf{R}(S) - \mathbf{E}_{S \sim \mathcal{X}^m}[\mathbf{R}(S)] \geq -\sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \geq 1 - \frac{\delta}{4}, \quad (23)$$

With equation 20 and 23,

$$\Pr[L(\hat{h}) - L_S(\hat{h}) - 2\mathbf{R}(S) \leq 3\sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \geq 1 - \frac{\delta}{2}, \quad (24)$$

Also remember $L(\hat{h}) \leq L(h^*) + \epsilon$, $L_S(\hat{h}) \geq L_S(h^*)$ in (a).

$$\Pr[L(\hat{h}) - L_S(h^*) - 2\mathbf{R}(S) \leq 3\sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \geq 1 - \frac{\delta}{2}, \quad (25)$$

since we are still one more step from success, we additionally define $X_i \triangleq \mathbf{E}[L_h(S^*) \mid S_0 \dots S_i]$. $\mathbf{E}[L_S(h^*)] = L(h^*)$. Alike equation 23 and 18 we have

$$\Pr[L_S(h^*) - L(h^*) \leq \sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \geq 1 - \frac{\delta}{4}, \quad (26)$$

Finally,

$$\Pr[L_S(\hat{h}) \leq L(h^*) + 2\mathbf{R}(S) + 4\sqrt{\frac{1}{2m} \log \frac{8}{\delta}}] \geq 1 - \frac{3\delta}{4} \quad (27)$$

References

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